CRISP-DM Report: Short-Term Stock Portfolio Optimization for Retail Investors

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I Business Understanding

A. Determine Business Objectives

Background: The project is initiated by an investment advisory business unit that is responsible for supporting retail clients in making stock market decisions. This unit defines business objectives, evaluates the usefulness of the results, and ensures alignment between analytical outcomes and real-world investment needs. Within the project, it acts as the business customer.

The problem area addressed in this project is short-term investment decision support. Retail investors often lack data-driven tools for selecting stocks that balance return and risk within a limited investment horizon. Existing approaches are typically based on manual analysis or simple heuristics, which may not fully exploit historical market data and can lead to suboptimal investment decisions.

The primary target user group consists of private investors with a six-month investment horizon, aiming to preserve and moderately increase their savings for future expenses. Secondary users include business analysts and investment specialists, who interpret model outputs and translate them into actionable recommendations.

From a business perspective, users expect the project to identify stocks with growth potential, provide interpretable and transparent reasoning behind selections, and support risk-aware investment strategies. In particular, given the short investment horizon, the solution is expected to help manage downside risk while achieving stable returns.

Business Objective: The objective of this project is to support short-term investment decision-making by identifying a portfolio of stocks that can provide positive returns over a six-month investment horizon. From a business perspective, the goal is to achieve a stable increase in the investor's deposit while maintaining an acceptable level of risk.

To make the objective measurable, the target return is set to at least **11.2%** over the investment period. This value is motivated by the need to outperform inflation by a significant margin: the official inflation rate in Russia was 5.6% in 2025 (Bank of Russia), and the project aims to achieve at least twice this level to ensure real growth of purchasing power.

At the same time, risk management constraints are introduced: the maximum acceptable loss for an investment strategy is limited to **20%** of the initial deposit. This threshold reflects a moderate risk tolerance for retail investors and is supported by observed drawdowns in comparable investment strategies. For example, the "Rational Profile" strategy demonstrates a historical drawdown of approximately 23.38% (investment strategy description), which justifies setting a conservative threshold slightly above this level.

In addition, the solution is expected to provide interpretable recommendations that can be used by analysts and understood by end users. From a business value perspective, achieving these objectives allows investors to outperform inflation and traditional bank deposits, thereby increasing the attractiveness of stock market investments.

Business Success Criteria: The project will be considered successful if the proposed investment strategy achieves a return of at least **11.2%** over the six-month horizon while not exceeding the maximum acceptable loss of **20%**, as defined in the business objective.

B. Situation Assessment

Inventory of Resources:

- People: We have project manager, business unit, business analyst, data scientist, and machine learning engineer people in the team, so the team is complete for rigorous study.
- Data: we will parse 249 most popular Russian stocks, which are easily available from the Moscow Exchange Information and Statistical Service (MOEX ISS) API with different time frames ranging from minutes to months.
- Computing resources: Colab and Kaggle are great choices for programming and modeling.
- Software: Python with its numerous libraries and possibly Docker for deploying models.

Requirements, Assumptions, and Constraints:

- Requirements: historical stocks data for 16 years (15 years for modeling + 1 for evaluating real case scenario). The project duration is estimated to 2 months. Different time frames from minutes to months are needed for in-depth analysis of stocks behavior. The issue is that the Moscow Exchange requires people that are interested to profit from their data, need to contact the officials for negotiating commercial data use (User Agreement) (In Russian).

- Assumptions: the main assumption is that chosen historical period is enough to model profitable investment strategy. Hidden assumption could be that no serious crisis will happen during evaluation year, however this is addressed by that fact that some crises are already included in the data.

- Constraints: The project is subject to several practical constraints. First, the timeline is limited to approximately two months, as defined in the project plan, which restricts the complexity of data processing, feature engineering, and model development.

  Budget is also a constraint, since the project requires human resources, computing equipment, data access, and future promotion costs. Therefore, the solution should remain economically justified relative to the expected business value.

  Legal constraints include restrictions on the use of MOEX data, which may require compliance with licensing agreements and exchange data usage policies. In addition, if the system is later applied to real clients, personal data processing and user profiling must comply with privacy regulations such as GDPR or similar data protection standards.

  Technical constraints arise from limited computational resources, restricted project duration, and the need to process large volumes of historical financial data within reasonable time limits.

Risks and Contingencies: Several risks may affect the successful completion and outcomes of the project.

Data risks: The dataset may contain missing values, inconsistencies, or biases (e.g., survivorship bias or incomplete historical records). These issues can negatively impact model training and evaluation. To mitigate this risk, data quality analysis will be performed during the data understanding phase, and appropriate preprocessing techniques such as imputation, filtering, and validation against external sources will be applied.

Technical risks: The developed models may fail to achieve the desired predictive performance or may overfit historical data, leading to poor generalization on unseen data. To address this, multiple modeling approaches will be tested, cross-validation techniques will be applied, and model performance will be continuously evaluated against defined success criteria.

Business risks: The resulting investment strategy may not be adopted by end users due to lack of trust, insufficient interpretability, or misalignment with user expectations. To reduce this risk, emphasis will be placed on model interpretability and clear presentation of results, ensuring that recommendations are understandable and actionable for both analysts and investors.

External risks: Unexpected market conditions, such as financial crises or extreme volatility, may invalidate patterns learned from historical data. As a contingency, risk management rules (e.g., stop-loss constraints) are incorporated, and the limitations of the model will be explicitly communicated.

Terminology:

- Stock - partial ownership in an organization.
- Dividend - a sum of money paid by a company to its shareholders out of its profits.

Costs and Benefits: **Costs:** The economic feasibility of the project can be assessed by comparing the total implementation and promotion costs with the expected business benefits.

The project cost consists of several components. First, labor costs are estimated using an average junior IT specialist salary of \$1000 per month. For a team of five members working for two months, the total labor cost is

$$5 \times 1000 \times 2 = \$10{,}000.$$

Second, infrastructure costs are estimated through laptop depreciation. Assuming that each team member uses a laptop costing \$2000 with a renewal period of four years, the monthly cost per laptop is

$$2000/48 \approx \$41.7.$$

For five laptops over two months, the total infrastructure cost is

$$41.7 \times 5 \times 2 \approx \$417.$$

Third, data acquisition costs are associated with MOEX market data access. Assuming an average cost of 20,000 RUB per month and an exchange rate of 1 USD = 80 RUB, the monthly cost is

$$20000/80 = \$250.$$

For two months, this gives

$$250 \times 2 = \$500.$$

Fourth, operational costs such as electricity, internet access, and auxiliary software usage are estimated at approximately \$100 per month for the whole team, resulting in

$$100 \times 2 = \$200.$$

In addition, promotion costs should be considered for the future deployment of the solution. Assuming a marketing campaign of 5 paid posts per day in different channels, with an average cost of \$10 per post, the daily marketing cost is

$$5 \times 10 = \$50.$$

For three months of promotion (approximately 90 days), the total marketing cost is

$$50 \times 90 = \$4{,}500.$$

Thus, the total estimated cost of the project and initial promotion stage is

$$10{,}000 + 417 + 500 + 200 + 4{,}500 = \$15{,}617.$$

Benefits: The expected benefits of the project arise from attracting investors to a strategy that aims to outperform inflation and traditional bank deposits. If the solution is adopted by clients, it can generate business value through increased customer engagement, higher trust in investment recommendations, and potential revenue from advisory or portfolio management services. Therefore, the project becomes economically justified if the resulting strategy attracts enough users or managed capital to compensate for the estimated total cost of \$15,617.

The expected benefits of the project can be quantified by modeling potential revenue from providing investment recommendations. Let us assume that the business model is based on charging a commission of **5%** from the positive return generated for each client.

Assuming an average client investment of \$5,000 and a target return of 11.2%, the expected profit per client over the six-month horizon is

$$5000 \times 0.112 = \$560.$$

The corresponding revenue for the business (5% commission) is

$$560 \times 0.05 = \$28 \text{ per client.}$$

To cover the total project and promotion cost of \$15,617, the required number of clients is

$$15617/28 \approx 558 \text{ clients.}$$

Thus, the project reaches the break-even point when approximately **560 clients** adopt the strategy.

If the number of clients exceeds this threshold, the project becomes profitable. For example, with 1000 clients, the expected revenue is

$$1000 \times 28 = \$28{,}000,$$

resulting in a net profit of

$$28000 - 15617 = \$12{,}383.$$

In terms of payback period, assuming a steady inflow of clients and that each client completes one investment cycle every six months, the project is expected to recover its initial costs within the first cycle if at least 560 clients are acquired. Otherwise, the payback period extends proportionally (e.g., approximately one year if only half of the required clients are reached in the first cycle).

Therefore, the project becomes economically viable if it can attract a moderate number of users, demonstrating that even a relatively small client base can compensate for the initial investment and generate profit.

C. Determine Data Mining Goals

Data Mining Goals: The business objective of achieving stable positive returns with controlled risk is translated into a data mining task focused on predicting stock performance over a six-month horizon.

The primary data mining problem is defined as a **regression task**, where the goal is to predict the future return of a stock based on historical price data and relevant features. This can be formulated as a time series forecasting problem, where past observations are used to estimate future price movements.

The output of the data mining process is a model that assigns a predicted return (or probability of profitability) to each stock, allowing the construction of an investment portfolio that aligns with the business objective.

Data Mining Success Criteria: The performance of the model will be evaluated using technical metrics appropriate for the defined problem types. For the regression task, the primary evaluation metric is **Mean Absolute Error (MAE)**, which measures the average deviation between predicted and actual returns. Lower MAE values indicate more accurate predictions.

To ensure practical usability, the model must also satisfy interpretability requirements, allowing analysts to understand the key factors influencing predictions. This is important to support trust and adoption of the model in real-world decision-making.

D. Produce Project Plan

Phase	Duration	Resources	Inputs	Outputs	Dependencies
Business Understanding	0.5 Weeks	Business Analyst, Project Manager, Business Unit	Business Requirements, Domain Knowledge	Business Objectives, Situation Assessment, Success Criteria, Project Plan	None
Data Understanding	1.5 Weeks	Data Scientist, Business Analyst, Project Manager	MOEX Stocks Parsed Data	Initial Data Collection Report, Data Description Report, Data Exploration Report, Data Quality Report	Business Understanding
Data Preparation	1.5 Weeks	Data Scientist, Business Analyst, Project Manager	Raw Data with Description, Exploration, Quality Descriptions	Data Set(s) with Description	Data Understanding
Modeling (Iteration 1)	1 Week	ML Engineer, Project Manager	Prepared Dataset(s)	Baseline Models, Initial Predictions	Data Preparation
Evaluation (Iteration 1)	1 Week	Data Scientist, Business Analyst, Business Unit, Project Manager	Baseline Model(s), Test Data	Model Performance Metrics, Business Objectives Alignment Report	Modeling (Iteration 1)
Modeling (Iteration 2)	1 Week	ML Engineer, Project Manager	Evaluation Feedback, Refined Data	Improved Model(s), Hyperparameter Tuning Results	Evaluation (Iteration 1)
Evaluation (Iteration 2)	1 Week	Data Scientist, Business Analyst, Business Unit, Project Manager	Improved Model(s)	Final Model Selection, Deployment Readiness Report	Modeling (Iteration 2)
Deployment	0.5 Weeks	Data Scientist, ML Engineer, Business Analyst	Final Model, Infrastructure Tools	Production System, Monitoring Dashboard and Logs	Evaluation (Iteration 2)

Table I: Project Plan

Project Plan: Estimated total project duration is 7-8 weeks (note that 8th week, which is dedicated to deployment, is under discussion) as shown in Table I

2 modeling iterations are planned as an optimal setup to try twice and analyze what went well and what did not.

Risk	**Action**	**Schedule Reflection**
Serious Data Quality Issues (Too Many Missing Values and/or Anomalies)	Prepare Data Imputation/Dropping Strategies	Extends Data Preparation for 1 week
Insufficient Expressiveness of Features	Collect More Data from Other Sources	Extends Modeling for 1 week and possibly Data Preparation for 1 week (Backtrack to Previous Stages if Needed)
Key People Unavailability	Documentation and involving specialists with similar background	Depends on situation, timeline extension for several weeks
Global Market Changes	Be Flexible and not Reactive to Changes, Understand Shifts First	Freezing the Project or Serious Timeline Extension

Table II: Risk Analysis with Schedule Impact and Actions

Risk Analysis and Schedule Dependencies: The timeline is presented in the Project Plan Table I. Data Preparation is the most risky phase for delays due to typical data quality and transformations challenges. This and other risks are detailed in the Table II

Review Points: Review points are scheduled as follows:

- **Review 1 (Week 0.5):** Business Understanding completion - verify match of data mining goals with business objectives before any work begins.
- **Review 2 (Week 2):** Data Understanding completion - assess data quality. Decide whether to proceed or find additional data.
- **Review 3 (Week 3.5):** Data Preparation completion - validate prepared dataset(s).
- **Review 4 (Week 5.5):** Iteration 1 Evaluation - compare initial business objectives with the results. Go to Business Understanding if mismatch is found.
- **Review 5 (Week 7.5):** Iteration 2 Evaluation - final model(s) selection and preparing for deployment.

- **Review 6 (Week 8):** Deployment completion - monitoring model(s) behavior in real world.

Initial Assessment of Tools and Techniques: Python has a plenty of great libraries with maintenance, so problems with data processing or modeling tools absence are not likely. Docker is a good deployment software as practice shows.

Trial of simple models first (time series decomposition) seems relatively efficient and explainable from business perspective, and neural networks may enhance the quality of the final result.

II Data Understanding

A. Data Selection

Before collecting any data, we decided which stocks, time periods, and tables to use based on the business goal:

The dataset includes ordinary shares of the 249 most traded Russian companies on the Moscow Exchange. The number 249 comes from the Broad Market Index (IMOEX) plus the most liquid securities from the second and third trading levels, as defined by the Exchange. This number balances variety and data quality. Fewer stocks, such as the top 50 only, would limit choices and increase risk from relying on too few companies. More stocks, such as all 800 available, would include many rarely traded instruments with too many missing values and unreliable prices. We selected only ordinary shares because they carry voting rights and dividends, which align with our total-return investment goal. We excluded preferred shares and depositary receipts to avoid introducing inconsistencies in dividend mechanics and trading frequency, which could add noise to the model and complicate the calculation of a uniform target variable.

The collected data covers 16 years, from 1 February 2010 to 1 February 2026. We chose this period because it includes different market situations: the 2014 sanctions, the 2020 pandemic crash, and the 2022 volatility. The last year (2025–2026) is kept aside for testing on unseen data. A shorter period would risk the model learning only one type of market condition. A longer period would include price behavior from before 2008 that no longer reflects today's market. Therefore, we excluded all data before 2010.

We selected four types of tables. The Market Candles tables give price data at six intervals: 10 minutes, 1 hour, 1 day, 1 week, 1 month, and 1 quarter. Short intervals help us understand price changes within a day and choose good entry points. Longer intervals show clear trends over weeks and months. The Market History table provides daily open, high, low, close, and trading volume. This data is needed to calculate returns and losses over the one-year period. For both Market Candles and Market History, we included data from all trading modes because limiting to a single mode would exclude many trades and reduce data completeness. The Dividends table records past dividend payments per share, including the date when shareholders must be registered and the currency used. This allows us to calculate total return, not just price change. The Security Descriptions table gives fixed information about each stock, such as industry sector, listing level, and high-risk flags. This helps us filter stocks by company characteristics.

B. Data Collection

The dataset was collected by parsing historical market data from the Moscow Exchange Information and Statistical Service (MOEX ISS) API. The data collection was implemented using the `apimoex` Python library, which is convenient for accessing MOEX market data endpoints. The collected data covers the period from `01.02.2010` to `01.02.2026` and includes daily trading records, historical dividend payments, candlestick data across multiple time intervals, and securities description for the predefined list of 249 most popular Russian stocks.

C. Dataset Description

The dataset is organized into four table types:

Market Candles Tables: These tables represent historical candlestick data aggregated across multiple time intervals for all securities and trading boards. The data is stored in a hierarchical folder structure, with separate subfolders for each interval: `10min`, `1hour`, `1day`, `1week`, `1month`, and `1quarter`. Within each interval folder, individual CSV files follow the naming pattern {ticker}-{interval}.csv. The CSV schema includes the candle start time (begin), the four standard price points (open, high, low, close), the total trading value in rubles (value), and the volume in lots (volume).

Dividends Table (dividends.csv): This table captures the historical dividend payments for securities in the dataset. Each record contains the ticker (secid), the international identifier (isin), the critical register closure date (registryclosedate), the dividend amount per share (value), and the currency of payment (currencyid).

Market History Table (market_history.csv): This table contains daily trading records for all securities across all trading boards. Each row is uniquely identified by the composite key of ticker (SECID), trading board identifier (BOARDID), and trading date (TRADEDATE). The CSV schema includes opening price (OPEN), intraday peak (HIGH), intraday trough (LOW), and closing price (CLOSE), along with trading volume in lots (VOLUME) and total monetary value of all trades (VALUE). Additional metrics include the volume-weighted average price (WAPRICE), the number of individual transactions (NUMTRADES), the denomination currency (CURRENCYID), security's short name (SHORTNAME), official settlement closing price (LEGALCLOSEPRICE), trend closing price (TRENDCLSPR), market prices (MARKETPRICE2, MARKETPRICE3), admitted quote (ADMITTEDQUOTE), associated trade values (MP2VALTRD, MARKETPRICE3TRADESVALUE, ADMITTEDVALUE), weighted average value (WAVAL), trading session identifier (TRADINGSESSION), and trade session date (TRADE_SESSION_DATE).

Security Descriptions Table (security_descriptions.csv): This table acts as a static reference, containing one row per ticker with unchanging metadata about each security. Key columns include the ticker (SECID), the issuer's full name (NAME), the trading short name (SHORTNAME), the international identifier (ISIN), the state registration number (REGNUMBER), the nominal value (FACEVALUE) in the specified unit (FACEUNIT), the total issue size (ISSUESIZE), the issue date (ISSUEDATE), the listing level (LISTLEVEL), and classification fields (GROUP, TYPE, TYPENAME). Additional metadata covers the emitter identifier (EMITTER_ID), the registry date (REGISTRY_DATE), trading session flags (MORNINGSESSION, EVENINGSESSION, WEEKENDSESSION), qualified investor status (ISQUALIFIEDINVESTORS, QUALINVESTORGROUP), high-risk flag (HIGHRISK), and depositary organization details (`DEPOSITARY_ORG_NAME`, `DEPOSITARYNOTETYPE`).

D. Data Exploration

The data exploration concentrated on the primary analytical table: the 1-day Market Candles dataset. After parsing and concatenating the individual security files, this dataset comprised 737,553 records covering 247 unique securities. The observation period spanned from 1 February 2010 to 1 February 2026, providing a comprehensive temporal foundation for subsequent analysis.

Temporal Analysis of Data Coverage: A first step was to assess the completeness of each security's price history. Insufficient historical depth can lead to unstable calculations for long-horizon features, which are essential for the predictive modeling objectives of this project. To investigate this, the number of years of available data was computed for each security and visualized through a histogram, as shown in Figure 1.

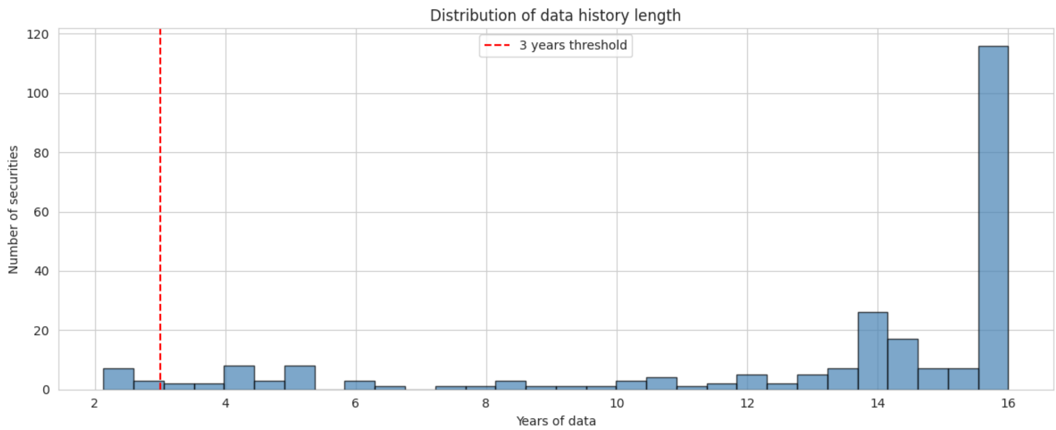

Figure 1: Distribution of data history length across securities. The red dashed line indicates the 3-year minimum threshold.

The analysis of data coverage, visualized in Figure 1, reveals a wide distribution of history length across securities. Crucially, 237 of 247 securities possess more than three years of data.

This finding directly motivates a key data preparation rule: a 3-year minimum history threshold. This threshold is not arbitrary; it is the minimum lookback period required to calculate our longest-horizon feature, the 36-month momentum, without introducing instability from insufficient data. Securities failing to meet this criterion will be systematically excluded to ensure that all features in the modeling dataset are computed on a robust and comparable basis, thereby avoiding a potential source of noise in model training.

Liquidity Analysis: Liquidity, defined as the ease of trading an asset without causing a significant price impact, is a critical constraint for portfolio feasibility. In this analysis, it is quantified as the average daily trading value in Russian Rubles (RUB). A strategy, regardless of its theoretical performance, has no practical value if its recommended securities cannot be traded efficiently.

To reflect the current market microstructure, the analysis focused on trading activity during the most recent full year (2024). Figure 2 presents the top 30 securities by this liquidity measure.

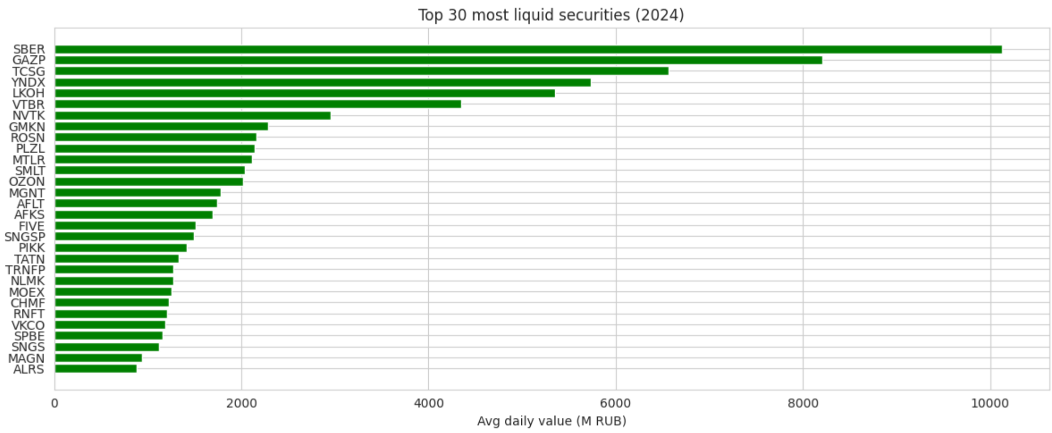

Figure 2: Top 30 most liquid securities based on average daily trading value (RUB) in 2024.

The bar chart in Figure 2 reveals a highly concentrated liquidity profile, a characteristic pattern observed in the Russian equity market. A small cohort of premier stocks dominates trading volumes, while a long tail of securities demonstrates structurally lower levels of activity.

This finding has direct implications for the portfolio construction phase: predicted returns alone are an insufficient selection criterion. To ensure the practical investability of the final portfolio, a liquidity-aware filter must be integrated. Based on this analysis, a minimum threshold of 1 million RUB in average daily trading value will be applied. This rule retains 215 securities, defining a sufficiently broad yet adequately liquid investment universe for the subsequent modeling stage.

Dividends and Delisting Overview: Understanding corporate actions, specifically dividend payments and listing status, is fundamental for accurately computing total shareholder return and avoiding systematic biases in model training.

Active vs. Delisted Securities: An analysis of security status revealed that of the 247 unique securities, 231 remained actively trading as of 1 January 2025, while 16 had been delisted. The presence of delisted securities introduces a significant risk of survivorship bias. If left unaddressed, the model would be trained exclusively on securities that survived the entire period, implicitly learning patterns associated with success and failing to generalize to stocks that may underperform and be removed from the exchange. This concern will be a central consideration during data preprocessing and model validation, potentially necessitating the use of point-in-time data construction to prevent look-ahead bias.

Dividend Coverage: A total of 163 unique securities had paid at least one dividend over the sample period. The number of dividend-paying companies has shown a generally upward trend, confirming the increasing importance of dividends as a component of total return in the Russian market.

The growing prevalence of dividend payments implies that an analysis based purely on price dynamics would systematically ignore an increasingly significant driver of shareholder value, especially for income-oriented strategies. This finding presents a compelling opportunity to develop features related to dividend yield and payment consistency, which are expected to enhance the predictive power of the final model.

E. Data Quality Verification

The quality and integrity of the core `market_history` dataset were subjected to rigorous verification immediately upon loading and concatenation of the individual CSV files. This step aimed to identify any data anomalies early in the project lifecycle, before they could propagate through the feature engineering and modeling stages.

Missing Values: An initial check for missing values was performed on the critical price and volume columns: `OPEN`, `HIGH`, `LOW`, `CLOSE`, `VOLUME`, and `VALUE`.

The check returned precisely 0.0% missing values across all key columns, indicating an exceptional level of completeness in the raw MOEX data. This finding positively resolves the “Data Risks” identified in the Business Understanding phase, specifically the risk of extensive missing values, and confirms that no complex imputation strategies are required at the start of the Data Preparation phase.

Logical Integrity: Beyond completeness, the logical integrity of the price data was verified by testing fundamental internal consistency rules:

- The `LOW` price must not be greater than the `OPEN`, `CLOSE`, or `HIGH` prices.
- The `HIGH` price must not be less than the `OPEN`, `CLOSE`, or `LOW` prices.

The data exhibited perfect logical integrity, with no violations of the basic low/high price relationships detected. This reliability confirms that no manual cleaning or corrective measures are required to address such anomalies, allowing the data to be used directly for analysis and feature generation.

III Data Preparation

The Data Preparation phase transformed the validated raw dataset into a clean, integrated, and feature-rich analytical table suitable for predictive modeling. All decisions were guided by findings from the Data Understanding phase.

A. Select Data

The primary source was the 1-day Market Candles dataset, loaded from the hierarchical folder structure and standardized by renaming columns to a consistent schema (`TRADEDATE`, `OPEN`, `HIGH`, `LOW`, `CLOSE`, `VOLUME`, `VALUE`) with a `SECID` column derived from each filename. The initial dataset contained 737,553 records for 247 securities, spanning 1 February 2010 to 1 February 2026.

All 247 securities were retained at this stage, including the 16 delisted by January 2025. Excluding delisted securities would introduce survivorship bias, as the model would only observe patterns from surviving companies. By retaining their full historical data, the model can learn from both successful and unsuccessful outcomes.

The 3-year minimum history threshold identified during Data Understanding was deferred to the end of the pipeline. Applying it prematurely would discard observations before features requiring long lookback windows could be computed.

B. Clean Data

The Data Quality verification confirmed 0.0% missing values and perfect logical integrity in the raw data, so extensive cleaning was unnecessary. The focus was on structural organization and outlier mitigation.

The dataset was sorted by `SECID` and `TRADEDATE` to guarantee chronological order for all time-dependent feature calculations. Rows with missing core price values were dropped as a precautionary measure; consistent with prior findings, no observations were removed.

Outlier Mitigation: Engineered features were clipped to reduce the influence of extreme values on model training. For each feature, values below the 0.1st percentile or above the 99.9th percentile were replaced with `NaN`. The 0.1st and 99.9th percentiles were chosen to remove only the most extreme observations (0.2% of the data) while preserving the vast majority of the distribution. The static `log_issuesize` feature was exempted because it is time-invariant for each security and represents a structural company characteristic rather than a market-driven metric. The `beta_12m` feature was additionally bounded to $[-5, 5]$ to limit the influence of unstable regression estimates that occasionally produce extreme values when the market index variance approaches zero.

Target variables were separately clipped to remove implausible returns. Upper bounds of 10.0 (182-day), 5.0 (42-day), and 2.0 (14-day) were applied, reflecting the decreasing likelihood of extreme returns over shorter horizons. Returns below -0.99 (a greater than 99% total loss) were also removed across all targets, as such values are indicative of data errors or extraordinary corporate events rather than normal market behavior.

C. Construct Data

Feature engineering was organized into four groups: technical, financial, dividend, and static. The complete set of constructed columns with their business rationale is presented in Table III.

Technical Features capture price momentum, volatility, and downside risk across four lookback horizons (6, 12, 24, and 36 months, with each month approximated as 21 trading days). The four-horizon design allows the model to potentially learn patterns at different time scales and detect regime shifts. Momentum was calculated as $\frac{\mathrm{CLOSE}_t}{\mathrm{CLOSE}_{t-k}} - 1$. Volatility was computed as the annualized rolling standard deviation of daily returns, with annualization achieved by multiplying the rolling standard deviation by $\sqrt{252}$, the typical number of trading days in a year. Maximum drawdown captured the worst peak-to-trough decline within each window, which is directly relevant to the 20% maximum acceptable loss constraint defined in the business objective.

Financial Features measure market behavior and systematic risk exposure. Average trading value (`avg_value`) over 6, 12, and 36-month windows provides a time-varying liquidity proxy that updates as market conditions change, unlike the static analysis performed during Data Understanding. Relative strength was calculated as the ratio of the security's closing price to the MOEX index, with values above 1 indicating outperformance. Excess return measures cumulative outperformance over a 6-month window by compounding daily excess returns. Beta quantifies systematic market risk over a 12-month rolling window as the covariance of security returns with index returns divided by the variance of index returns.

Dividend Features were motivated by the exploration finding that 163 of 247 securities paid dividends, representing a growing component of total return. Raw dividend events from `dividends.csv` were aggregated to a monthly level. The 12-month dividend yield sums trailing dividends divided by current price, providing a direct income return measure. The 3-year average yield smooths this measure over a longer window to reduce sensitivity to one-time payments. Dividend consistency scores the fraction of the past three years with at least one payment, ranging from 0 (no dividends) to 1 (paid every year). For securities with no dividend history, all three features were set to `NaN`.

Static Feature: The natural logarithm of issue size (`log_issuesize`) from `security_descriptions.csv` provides a proxy for company scale. The logarithmic transformation compresses the wide range of issue sizes across companies, reducing skewness.

Target Variables: The forward total return incorporates both price appreciation and dividends, aligning with the business objective of maximizing total shareholder return. For each observation, the future price was identified as the closing price on the trading day closest to the target horizon, and intervening dividends were summed. The formula used is:

$$\text{Target Return} = \frac{\text{Future Price} - \text{Current Price} + \text{Dividends Paid}}{\text{Current Price}}$$

Three horizons were constructed: 14 days for very short-term signals, 42 days as an intermediate horizon, and 182 days (approximately 6 months) as the primary horizon aligned with the business objective.

Exploratory Analysis of Constructed Features: Following construction, the engineered features were examined to validate their properties and assess their relationship with the target variable. The 6-month forward return distribution is wide and right-skewed, indicating that models should not assume normally distributed targets. Spearman rank correlations between each feature and the target are modest, with no single variable dominating. Decile profile analysis reveals that relationships are generally non-monotonic: higher momentum does not consistently deliver higher returns, and the effect varies across volatility, beta, and liquidity regimes. These findings support the selection of non-linear ensemble models such as gradient boosting, which can capture interaction effects without requiring explicit specification.

D. Integrate Data

All features, targets, and external data sources were combined through sequential merge operations on the base Market Candles table.

Technical and financial features were computed directly on the base table. Financial features required isolating the MOEX index rows, calculating index-level metrics, and broadcasting values back to all securities by trading date. Dividend features followed a separate path: monthly aggregation of `dividends.csv`, security-by-security trailing sum computation, and merge on `SECID` and `TRADEDATE`. Targets were calculated independently by iterating over securities, identifying the closest future trading date, and summing intervening dividends.

The static `log_issuesize` was merged on `SECID`. An external CBR key rate history was integrated by mapping each trading date to the prevailing interest rate, providing macroeconomic context.

After integration, any row with a missing value in any feature or target column was dropped. This step is necessary because the planned modeling algorithms require complete feature vectors and cannot natively handle missing values. The 3-year minimum history threshold was then enforced. This filter was deliberately applied at the end of the pipeline, rather than at the beginning, to ensure it operates on the final feature set after all cleaning and integration steps are complete.

The reduction from the initial 247 securities to 143 in the final dataset is driven primarily by the requirement for a complete, dense feature matrix. Long-horizon rolling features produce `NaN` values for early observations of every security, and dividend features cannot be computed for the 84 securities that never paid a dividend. These are consequences of feature engineering requirements, not data quality problems, and are consistent with the Data Quality findings that the raw source data contains no missing values.

E. Format Data

The final dataset was structured as a single tidy table in "long" format with respect to time, where each row represents a unique security-date observation and each column represents a distinct feature or target variable. This format is the standard input structure for the cross-sectional machine learning models (scikit-learn, LightGBM, CatBoost) planned for the Modeling phase.

The dataset contains 274,534 rows and 30 columns, spanning 9 February 2016 to 31 January 2026. The start date is later than the raw data because long-horizon features require several years of prior observations before they can be computed. The dataset was saved as `final_dataset.csv` and requires no further transformation before use.

Feature	Type	Description and Rationale
SECID, TRADEDATE	Identifier	Unique security and observation date.
year_month	Temporal	Aggregated month indicator for temporal filtering.
CLOSE	Price	Closing price used as base for target calculation.
momentum_6m	Technical	6-month price return; short-term trend signal.
momentum_12m	Technical	12-month price return; medium-term trend signal.
momentum_24m	Technical	24-month price return; long-term trend signal.
momentum_36m	Technical	36-month price return; multi-year trend signal.
volatility_6m	Technical	6-month annualized volatility; short-term risk.
volatility_12m	Technical	12-month annualized volatility; medium-term risk.
volatility_24m	Technical	24-month annualized volatility; long-term risk.
volatility_36m	Technical	36-month annualized volatility; multi-year risk.
max_drawdown_6m	Technical	6-month maximum drawdown; short-term downside risk.
max_drawdown_12m	Technical	12-month maximum drawdown; relates to 20% loss constraint.
max_drawdown_24m	Technical	24-month maximum drawdown; longer-term downside risk.
max_drawdown_36m	Technical	36-month maximum drawdown; multi-year downside risk.
avg_value_6m	Financial	6-month mean daily trading value; recent liquidity.
avg_value_12m	Financial	12-month mean daily trading value; medium-term liquidity.
avg_value_36m	Financial	36-month mean daily trading value; long-term liquidity.
relative_strength_12m	Financial	Price ratio to MOEX index; relative performance.
excess_return_6m	Financial	6-month cumulative excess return over MOEX index.
beta_12m	Financial	12-month rolling market beta; systematic risk exposure.
dividend_yield_12m	Dividend	Trailing 12-month dividend yield; income return.
dividend_yield_3y_avg	Dividend	Average annual yield over 3 years; smoothed income measure.
dividend_consistency_3y	Dividend	Fraction of past 3 years with dividend; payment reliability (0–1).
log_issuesize	Static	Log of total issued shares; company size proxy.
cbr_key_rate	Macro	Prevailing CBR key interest rate; macroeconomic context.
target_return_14d	Target	14-day forward total return (price + dividends).
target_return_42d	Target	42-day forward total return; intermediate horizon.
target_return_182d	Target	182-day forward total return; primary 6-month target.

Table III: Features in the final modeling dataset with descriptions and business rationale.

IV Modeling

The Modeling phase consisted of two iterations, reflecting an evolution in our modeling strategy. Iteration 1 focused on a single target variable with a single 180-day prediction horizon, establishing a baseline. Iteration 2 introduced a multi-horizon ensemble approach using three horizons (14, 42, and 182 days), designed to capture multi-scale market dynamics and improve predictive accuracy. This section details the technical rationale, design, and comparative assessment of these models.

A. Select Modeling Technique

The data mining goal is a regression task: predicting the forward total return of a stock. The exploratory analysis of constructed features during Data Preparation revealed that relationships between features and the target are generally non-monotonic and likely involve complex interactions. Therefore, linear models were considered insufficient.

For both iterations, we selected **XGBoost** (eXtreme Gradient Boosting) as the core modeling technique. XGBoost is a tree-based ensemble learning algorithm based on gradient boosting. The decision to use XGBoost was driven by several factors:

- **Non-linearity:** As a decision-tree ensemble, it can capture complex non-linear relationships and interactions between features without requiring explicit specification. This is essential for financial data where the effect of momentum, for instance, may depend on volatility or liquidity regimes.

- **Robustness to Scaling:** XGBoost is not sensitive to the scale of input features. Although we applied standardization for consistency, the algorithm would perform identically on raw features.

- **Regularization:** It offers built-in L1 and L2 regularization, which helps prevent overfitting on financial data known for its low signal-to-noise ratio.

- **Performance:** XGBoost is known for its state-of-the-art performance on structured/tabular data, consistently outperforming other algorithms in financial prediction benchmarks.

Modeling Assumptions: The application of XGBoost to our problem carries the following assumptions:

1. **Feature predictive power:** The engineered features contain sufficient signal to forecast future returns above a naive baseline.

2. **Stationarity of relationships:** The patterns learned from historical data (2010–2024) remain valid for the forward test period (2024–2025), despite known regime shifts in the Russian market.

3. **Complete feature vectors:** All missing values were removed during Data Preparation, as XGBoost does not natively handle missing targets.

4. **Temporal dependence addressed:** Although financial time series violate the independence assumption, our walk-forward validation design (described in Generate Test Design) ensures that no future information leaks into training.

Modeling Strategy Evolution: Our approach evolved over two iterations:

Iteration 1: Single-Horizon Model. The initial strategy involved predicting a single, 180-day forward target return using one XGBoost model. This approach was a direct translation of the business objective into a modeling task. However, it did not leverage information from shorter-term price dynamics, which we hypothesized would improve predictive accuracy.

Iteration 2: Multi-Horizon Ensemble Model. In the second iteration, we decomposed the prediction problem into three distinct forward horizons: 14 days, 42 days, and 182 days. Separate XGBoost models were trained for each horizon. The final prediction is a weighted ensemble, with weights dynamically computed as the normalized inverse of each model's historical Mean Absolute Error (MAE). This gives more influence to horizons that demonstrate higher accuracy. The 182-day horizon was adjusted from Iteration 1's 180 days to align with the approximately 6-month investment period specified in the business objective.

B. Generate Test Design

To avoid look-ahead bias and overfitting in time-series prediction, we implemented a temporally-aware validation framework. The test design distinguishes three data roles: scaler fitting data, training/validation data, and a final hold-out test set.

Data Partitioning. The complete dataset spans from 2015-01-01 to 2025-06-06. Partitioning was performed strictly along the temporal axis:

- **Scaler fitting period:** 2015-01-01 to 2018-01-01. This three-year window was used exclusively to compute the mean and standard deviation for feature standardization, ensuring that scaling parameters are derived from early data only.
- **Training and validation period:** 2018-01-01 to 2024-01-01. Within this window, walk-forward validation was applied (described below).
- **Hold-out test period:** 2024-01-01 to 2025-06-06. This data was completely excluded from all training and hyperparameter decisions and was used only for final portfolio evaluation.

Walk-Forward Validation. Within the training period, an expanding window walk-forward strategy simulated real-world deployment. Starting from 2018-01-01, the model advanced in steps of 14 days. The 14-day step was chosen to match the shortest prediction horizon and to balance computational cost with sufficient validation granularity. At each step:

1. **Training window:** The model used all available data within a rolling 3-year lookback period from the current date. This ensures enough observations for stable training while adapting to recent market conditions.
2. **Prediction:** The trained model generated predictions for all securities on the current date.
3. **Error tracking:** Predictions were stored in a registry and later compared against realized returns once the respective horizon had elapsed. Errors were used to update ensemble weights dynamically.

This design realistically mimics live model operation, where models are updated with new data and assessed on forward-looking accuracy. It guarantees that no future information leaks into the training process at any point.

C. Build Model

Both models were built using the `XGBRegressor` from the `xgboost` library, wrapped in a custom `EnsembleOrchestrator` class that manages multi-horizon training, prediction, and error tracking.

Parameter Settings. Key hyperparameters were selected to balance model complexity with generalization. Initial values were chosen based on common practices for financial data (shallow trees, moderate learning rate) and refined through manual experimentation on walk-forward validation performance. The final configuration is presented in Table IV.

Parameter	Value	Rationale
`n_estimators`	300	Provides sufficient boosting iterations for complex pattern learning. The learning rate of 0.05 combined with 300 trees allows the model to fit gradually without overshooting.
`max_depth`	3	Restricts each tree to shallow interaction depth. Deeper trees (4–6) showed signs of overfitting during preliminary runs, where validation MAE stopped improving while training MAE continued decreasing.
`learning_rate`	0.05	A conservative step size that, together with 300 trees, produces a robust fit. Higher rates (0.1) converged faster but produced higher validation error.
`subsample`	0.8	Randomly samples 80% of training rows per tree, introducing stochasticity that reduces overfitting. Standard values (0.8–1.0) were tested; 0.8 provided the best validation performance.
`colsample_bytree`	0.8	Uses 80% of features per tree to decorrelate ensemble members and reduce variance.
`reg_lambda`	1.0	L2 regularization on leaf weights. The value 1.0 provides moderate penalty to prevent the model from fitting noise.
`random_state`	42	Fixed seed for reproducibility of all stochastic components.

Table IV: Key XGBoost model parameters and their rationale.

Model 1 (Iteration 1). A single `EnsembleOrchestrator` was configured with `horizons=[180]`. Training used a single target column `target_return`, representing the forward 180-day total return. The model's primary limitation is the reliance on a single long-horizon view, which makes predictions sensitive to noise accumulated over the 6-month period.

Model 2 (Iteration 2). The `EnsembleOrchestrator` was configured with `horizons=[14, 42, 182]`. For each horizon h, a separate XGBoost model was trained on the corresponding target variable `target_return_{h}d`. The final prediction P_{ensemble} for a security is computed as:

$$P_{\text{ensemble}} = \sum_{h \in \{14,42,182\}} w_h \cdot P_h$$

where $w_h = \frac{1/(\text{MAE}_h+\epsilon)}{\sum_k 1/(\text{MAE}_k+\epsilon)}$ and $\epsilon = 10^{-6}$ prevents division by zero. This weighting scheme automatically favors horizons with lower prediction error.

D. Assess Model

Model Comparison. The primary evaluation metric is MAE, measuring average absolute deviation of predicted returns from realized returns. Table V summarizes the results.

Model	**Horizon**	**Final MAE**	**Ensemble Weight**
Iteration 2	14 days	0.0670	60.7%
	42 days	0.2717	29.2%
	182 days	0.5734	10.1%
Iteration 1	180 days	0.8380	–

Table V: Model performance comparison. Lower MAE indicates better predictive accuracy.

The multi-horizon ensemble (Iteration 2) substantially outperforms the single-horizon baseline (Iteration 1). The 182-day model alone already improves on Iteration 1 (MAE 0.573 vs. 0.838), likely due to the cleaner multi-target dataset construction. The ensemble further benefits from the highly accurate short-term models: the 14-day model has an MAE of only 0.067, earning it approximately 60.7% of the ensemble weight. The 42-day model contributes 29.2%, while the 182-day model provides a 10.1% strategic long-term adjustment.

Ranking: Iteration 2 (Multi-Horizon Ensemble) is ranked superior to Iteration 1 (Single-Horizon) due to substantially lower prediction error and greater architectural robustness.

Revised Parameter Settings. Between Iteration 1 and Iteration 2, the following changes were made:

- **Horizon alignment:** The long horizon was changed from 180 to 182 days to better match the 6-month business objective (financial models operate on trading days rather than calendar days, and 182 trading days is the standard market proxy for a six-month forward period).

- **Target variables:** Iteration 1 used a generic `target_return` column. Iteration 2 introduced separate `target_return_14d`, `target_return_42d`, and `target_return_182d` columns with horizon-specific clipping thresholds.

- **Model hyperparameters:** The same XGBoost parameter settings were retained from Iteration 1, as they proved effective. The key improvement came from the architectural change (multi-horizon ensemble) rather than hyperparameter tuning.

Feature Importance. An analysis of feature importance for Model 2 was performed, with results visualized in Figure 3. The analysis accounts for the ensemble weight distribution: since the 14-day and 42-day models collectively represent approximately 90% of the final prediction, the feature importance profile naturally reflects the drivers of successful short-to-medium term forecasting, while still incorporating the long-term perspective from the 182-day model.

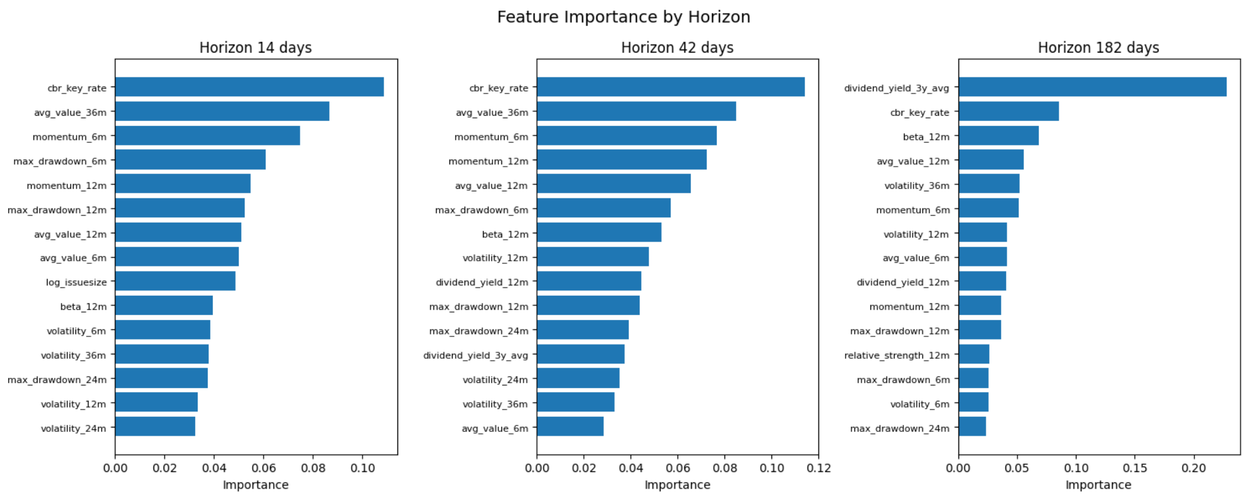

Figure 3: Feature importance scores for selected features across the 14-day, 42-day, and 182-day horizon models. Values represent the normalized contribution of each feature to the respective model's predictions.

Key findings from the feature importance analysis are:

- **Macroeconomic dominance:** The CBR key rate (`cbr_key_rate`) is the single most important predictor across all three horizons, with consistently the highest importance score. This aligns with financial intuition that monetary policy is a primary driver of equity market returns, particularly in the Russian market where interest rate changes have strong transmission effects.

- **Liquidity as a persistent signal:** Long-term average trading value (`avg_value_36m`) is the second most important feature across all horizons. This suggests that sustained trading activity over a three-year period captures fundamental company quality and market accessibility that short-term liquidity measures cannot.

- **Short-term technical indicators dominate price features:** Among price-derived features, `momentum_6m` and `max_drawdown_6m` are the strongest predictors. The model prioritizes recent trend and downside risk over longer-horizon price patterns, consistent with the finding that short-term models contribute most to ensemble accuracy.

- **Low-value features:** `relative_strength_12m` and `beta_12m` consistently rank near zero importance across all horizons. This may indicate that the model captures market-relative behavior internally through other feature combinations, rendering these explicit measures redundant.

Connection to Business Criteria. The ensemble model was evaluated on a forward-looking portfolio test spanning five dates from 2024-08-05 to 2025-06-06. The constructed portfolios achieved realized 182-day returns ranging from -0.8% to $+25.2\%$. While some test dates (e.g., 2024-09-03 with -0.8%) fell short of the 11.2% return target, the later test periods demonstrated strong positive performance. A comprehensive evaluation against the business success criteria, including risk metrics and consistency analysis, is presented in the Evaluation phase.

Overall, Model 2 satisfies the data mining success criteria of achieving low prediction error while maintaining interpretable feature relationships. The model's decisions are driven by economically meaningful factors - monetary policy, liquidity, and short-term price dynamics - rather than spurious correlations, supporting trust and adoption by analysts and investors.

V Evaluation

The Evaluation phase determines whether the results of the Modeling phase satisfy the business objectives and data mining goals defined in Section I.

A. Evaluate Results

Assessment of Data Mining Results with Respect to Business Success Criteria: The Business Success Criteria defined two measurable targets for the investment strategy over a six-month horizon: (1) achieve a return of at least 11.2%, and (2) not exceed a maximum loss of 20% of the initial deposit.

The final multi-horizon ensemble model was evaluated on a forward-looking portfolio test spanning five dates from 2024-08-05 to 2025-06-06. At each date, the model predicted returns for all available securities. These predictions were processed through the portfolio construction pipeline, which filters for positive predictions, applies HDBSCAN clustering for diversification, and weights selected stocks by prediction magnitude. The constructed portfolios were held for approximately six months, and realized total returns were calculated. Table VI presents the results.

Portfolio Date	Number of Stocks	Realized 182-Day Return
2024-08-05	8	+0.08%
2024-09-03	8	−0.82%
2025-04-03	18	+15.59%
2025-05-05	9	+25.16%
2025-06-06	9	+10.01%

Table VI: Realized portfolio returns for forward-looking test dates.

The results demonstrate partial success with respect to the business criteria. Three of the five test dates achieved returns near or above the 11.2% target: +15.59%, +25.16%, and +10.01%. The remaining two dates produced approximately break-even outcomes (+0.08% and −0.82%). Critically, no portfolio triggered the 20% maximum loss constraint at any test date. The worst observed loss was −0.82%, which is an order of magnitude below the risk limit.

Detailed portfolio compositions for all five test dates are provided in Appendix 0.6.1.

The variation in returns across test dates reflects the fundamental difficulty of predicting stock returns over a six-month horizon. The model identifies positive-return opportunities in favorable market conditions but cannot guarantee consistent outperformance. The portfolio clustering mechanism, which diversifies selections across market segments identified by HDBSCAN, contributes to the strategy's strong risk control by preventing over-concentration in any single segment. The realized returns are realistic in magnitude and direction: they are consistent with the well-documented variability of short-horizon stock prediction and do not suggest data leakage or overfitting.

The original business question asks whether a data-driven approach can identify portfolios with positive returns over a six-month horizon. The answer is **yes, partially**. The model successfully identifies such portfolios in the majority of test cases. However, it cannot guarantee positive returns in every period, which is an inherent limitation of financial prediction.

Final Statement on Business Objectives. The project **partially meets** the initial business objectives. The risk constraint of 20% maximum loss is satisfied with substantial margin. The return target of 11.2% is achieved in the majority of test cases but not consistently across all evaluation dates. The results are promising for a data-driven strategy but are not yet sufficiently robust to guarantee the target return to investors.

Assessment of Data Mining Results with Respect to Data Mining Success Criteria: The Data Mining Success Criteria specified two requirements: (1) low Mean Absolute Error for prediction accuracy, and (2) model interpretability to support analyst trust and adoption.

With respect to accuracy, the multi-horizon ensemble model achieved MAE values of 0.067 (14-day), 0.272 (42-day), and 0.573 (182-day). These represent a substantial improvement over the single-horizon baseline MAE of 0.838. While the 182-day error remains relatively high, this is consistent with the difficulty of long-horizon financial forecasting and does not prevent the model from constructing profitable portfolios, as demonstrated by the portfolio test results.

With respect to interpretability, the feature importance analysis (Section IV.D) revealed a clear hierarchy of economically meaningful predictors: the CBR key rate, long-term liquidity, and short-term momentum. The model's decisions are driven by factors that align with financial domain knowledge, and the negligible importance of certain features (relative strength, market beta) provides a clearer, simpler understanding of what drives returns. These characteristics satisfy the interpretability requirement and support the goal of building trust with analysts and end users.

The data mining results therefore **meet** the data mining success criteria: the model achieves meaningful predictive accuracy and produces interpretable outputs.

Approved Models: The multi-horizon ensemble model (Iteration 2) is approved as the project's final model. It substantially outperforms the single-horizon baseline (Iteration 1) and demonstrates the ability to construct portfolios with strong risk-adjusted returns. The model is approved for continued development and testing; it is not yet approved for deployment to real clients due to the inconsistency of returns identified in the business criteria assessment.

Additional Findings Not Directly Related to Business Objectives: Several findings emerged during the project that, while not directly measured by the business success criteria, provide valuable insights for future work:

- **Macroeconomic dominance.** The CBR key rate was the single most important predictor across all horizons. This implies that the model's recommendations will shift substantially when the central bank changes interest rates. End users should understand that portfolio recommendations reflect the prevailing monetary policy regime and may change materially after rate adjustments.
- **Redundancy of traditional indicators.** Relative strength and market beta, widely used in conventional financial analysis, contributed negligible predictive power. The model captures market-relative dynamics internally, rendering these handcrafted ratios unnecessary. Future feature engineering efforts could safely exclude them, reducing dataset dimensionality without performance loss.
- **Liquidity as a persistent filter.** Long-term average trading value was the second most important feature, confirming that sustained trading activity is a robust signal

of stock quality. This validates the decision, motivated by the Data Understanding liquidity analysis (Section II.D), to incorporate liquidity constraints in the portfolio construction pipeline.

B. Review Process

Quality Assurance: The modeling process incorporated several quality assurance measures. The walk-forward validation design prevented look-ahead bias by ensuring that all training data strictly preceded prediction dates. Feature standardization parameters were computed exclusively from the scaler fitting period (2015–2018), avoiding contamination from later data. The separate hold-out test period (2024–2025) was completely excluded from all modeling decisions. These procedures provide confidence that the reported results are not inflated by data leakage.

However, three quality assurance limitations must be noted. First, hyperparameter tuning was performed manually rather than through systematic grid search or cross-validation within the walk-forward framework. The selected parameters (Table IV) are reasonable and empirically justified, but the search space was not exhaustively explored. It is possible that a different configuration would yield better performance. The computational cost of walk-forward training (approximately 14 minutes per full run for Iteration 2) limited the number of configurations that could be tested within the project timeline.

Second, the five portfolio test dates, while covering different market conditions, represent a small evaluation sample. A larger number of evaluation points would be necessary to draw statistically robust conclusions about the strategy's consistency.

Third, data quality verification was performed thoroughly on the core market data table but was not formally documented for the dividends and security descriptions tables. While no issues were encountered in practice, this represents a minor gap in the quality assurance record.

Data Preparation Sufficiency: The data preparation choices proved sufficient for the modeling task. The decision to retain delisted securities in the training data, motivated by survivorship bias concerns in Section II.D, ensured that the model learned from both successful and unsuccessful stock histories. The outlier clipping thresholds (0.1st and 99.9th percentiles) successfully removed extreme values without discarding meaningful observations. The 3-year minimum history threshold, derived from the data coverage analysis in Section II.D, produced a stable feature matrix without excluding an excessive number of securities (247 reduced to 143, with the reduction primarily attributable to dividend feature requirements rather than the history threshold itself).

One limitation of the data preparation is the exclusion of 84 securities that never paid dividends during the sample period. These securities were dropped because dividend features cannot be computed for them. While methodologically necessary given the requirement for complete feature vectors, this exclusion means the model cannot evaluate non-dividend-paying stocks, which may include growth-oriented companies that could otherwise be attractive investment candidates.

Modeling Assumptions Revisited: The Modeling Assumptions stated in Section IV.A require re-examination in light of the results. The assumption of feature predictive power holds for short horizons but is weaker for the 182-day horizon, where MAE remains relatively high. The assumption of stationarity of relationships cannot be fully validated with only five test dates; the varying portfolio returns across test dates could indicate regime-dependent performance, which would violate this assumption. A longer evaluation period segmented by market conditions would be needed to test this properly. The assumptions of complete feature vectors and correct temporal handling were validated: no missing data issues arose during modeling, and the walk-forward design successfully prevented data leakage.

Process Strengths: Several aspects of the process were executed correctly:

- **Cross-phase traceability.** Decisions in later phases consistently referenced findings from earlier phases. For example, the selection of non-linear XGBoost was motivated by the exploratory finding that feature-target relationships are non-monotonic (Section III.C). The dividend features were directly motivated by the observation that 163 of 247 securities paid dividends (Section II.D). The liquidity constraints in portfolio construction originated from the liquidity concentration analysis (Section II.D). This traceability demonstrates disciplined application of the CRISP-DM methodology.

- **Iterative improvement.** The two-iteration modeling cycle allowed the team to diagnose the limitations of a single-horizon approach and implement a more sophisticated multi-horizon ensemble. The improvement from Iteration 1 to Iteration 2 confirms that the iterative structure added value.
- **Honest reporting.** Results are presented without cherry-picking. Both positive outcomes (+25.16%) and negative ones (−0.82%) are reported and discussed.

Process Weaknesses and Overlooked Tasks: Several tasks were insufficiently addressed or could be improved if the project were repeated:

- **No baseline model comparison.** XGBoost performance was not compared against simpler alternatives such as linear regression or a naive historical mean predictor. Without this comparison, it is impossible to quantify the marginal benefit of using a complex ensemble model. In a repeated project, at least one simple baseline would be implemented alongside XGBoost.
- **No formal statistical testing.** The conclusion that Iteration 2 outperforms Iteration 1 is based on the absolute difference in MAE values. A formal test for predictive accuracy differences (e.g., the Diebold-Mariano test) would provide stronger evidence that the improvement is statistically significant.
- **Unused initial technique ideas.** The Initial Assessment of Tools and Techniques (Section I.D) mentioned time series decomposition and neural networks as candidate methods. Time series decomposition proved unsuitable for the cross-sectional nature of the prediction task, while neural networks were not explored because of their lack of interpretability.
- **Insufficient evaluation sample.** The five-date portfolio test does not provide enough data points to calculate statistically meaningful performance metrics such as the Sharpe ratio or the probability of negative returns. If the project were repeated, the evaluation would be designed with a larger number of test dates from the outset.

Attribute Availability for Future Use: All features used in the final model are available for future analyses. Price-derived features can be recomputed from ongoing MOEX data feeds. The CBR key rate is publicly available from the Bank of Russia. Dividend data continues to be published by MOEX. Log issue size is static for each security and requires only one-time collection. The feature set is sustainable for ongoing deployment without dependency on proprietary or intermittent data sources.

C. Determine Next Steps

Based on the assessment of results and the review of the project process, three possible courses of action are identified:

1. **Proceed directly to deployment.** Deploy the current model to real clients.

   Arguments for: The model is functional and has demonstrated strong positive returns in the majority of test cases while maintaining losses well within the risk limit.

   Arguments against: Returns are inconsistent across test dates. Only five test dates have been evaluated. The risk of delivering near-zero or negative returns to clients in some periods is high. This option is premature given the current evidence.

2. **Conduct a third CRISP-DM iteration with targeted improvements.** Address the identified weaknesses, particularly the limited test evaluation and missing baseline comparisons, and reassess.

   Arguments for: The improvements are well-defined and would directly address the main limitation (inconsistent returns) by generating a statistically meaningful performance record. The resource cost is moderate (estimated 2–3 additional weeks). Additional data collection is not a priority for this iteration; the existing dataset is sufficient for the planned improvements. The feature set proved adequate, though removing zero-importance features (relative strength, market beta) could simplify the model without performance loss.

   Arguments against: The two-month project timeline has been exhausted. Additional time and budget would be required.

3. **Initiate a new data mining project with a different approach.** Redesign the problem, possibly changing the prediction task from regression to classification (profitable vs. unprofitable) or exploring deep learning methods.

   Arguments for: A fundamentally different approach might overcome the limitations of the current regression framework. Additional macroeconomic indicators beyond the CBR key rate—such as oil prices, exchange rates, or sector-specific indices—could be incorporated as features.

   Arguments against: The current approach has produced meaningful results. Starting from scratch would discard accumulated knowledge and require significant additional resources.

Decision: **Option 2: Conduct a third CRISP-DM iteration with targeted improvements**. The rationale is as follows:

The current model demonstrates a strong foundation: it controls risk effectively, it identifies profitable opportunities in favorable market conditions, it is interpretable through feature importance analysis, and it uses only data that is sustainably available. Its primary limitation - inconsistency of returns across evaluation dates - is addressable through the improvements identified in the process review.

The current result is not yet ready for deployment to real clients. The inconsistency of returns and the small evaluation sample mean that the strategy cannot be recommended to investors with sufficient confidence. A paper-trading or simulated deployment may be considered after the third iteration if the expanded evaluation demonstrates acceptable consistency.

The third iteration should prioritize the following actions:

1. **Expand the test evaluation.** Generate portfolio returns at monthly intervals over the entire 2024–2025 hold-out period, producing approximately 12–18 evaluation points. Calculate the mean return, standard deviation, hit rate, and maximum drawdown to assess consistency.

2. **Implement baseline models.** Add a naive historical mean predictor and a linear regression model to quantify the marginal value of XGBoost.

3. **Test robustness to portfolio construction rules.** Evaluate whether the HDBSCAN clustering step adds value over simpler selection rules such as equal-weighting the top-N predictions.

4. **Document performance in different market regimes.** Segment the evaluation period by CBR key rate trend (rising, falling, stable) to understand whether model performance is regime-dependent and to test the stationarity assumption.

If the expanded evaluation demonstrates a mean return near or above the 11.2% target with acceptable consistency and continued risk control, the project should proceed toward deployment. If the expanded evaluation reveals fundamental inconsistency, Option 3 should be reconsidered.

The team should also review the remaining budget and timeline with the business unit to secure resources for the additional iteration. The cost-benefit analysis in Section I.B indicates that even moderate client adoption (560 clients) covers project costs, providing economic justification for continued investment in model improvement.

VI Appendix

A. Portfolio Composition Details

This appendix presents the full composition of each portfolio constructed during the forward-looking evaluation. The tables list all selected stocks with their prediction scores, portfolio weights, HDBSCAN cluster assignments, and, where available, the realized 182-day target returns. A dash (—) in the Target Return column indicates that the realized return could not be determined because the security was delisted between the portfolio date and the horizon date or had no trading record on the relevant future date, and therefore no matching row existed in the dataset at the time of evaluation.

Table VII: Portfolio Composition (2024-08-05)

Ticker	**Prediction Score**	**Weight**	**Cluster**	**Target Return**
GMKN	0.5346	41.71%	-1	2.01%
TRNFP	0.2524	19.69%	-1	31.56%
RTSB	0.0995	7.77%	-1	-26.42%
MISBP	0.0862	6.72%	0	-11.83%
MISB	0.0857	6.69%	0	-13.66%
KCHE	0.0747	5.83%	0	—
ZVEZ	0.0782	6.10%	1	—
NSVZ	0.0703	5.49%	1	—

Portfolio Return: 0.08%

Table VIII: Portfolio Composition (2024-09-03)

Ticker	**Prediction Score**	**Weight**	**Cluster**	**Target Return**
TRNFP	0.2333	29.79%	-1	30.65%
NNSBP	0.1276	16.29%	0	-12.66%
IGSTP	0.0969	12.38%	-1	-28.24%
MISBP	0.0927	11.84%	0	1.33%
KCHE	0.0922	11.77%	0	-17.16%
RTSB	0.0880	11.24%	-1	—
USBN	0.0302	3.86%	1	—
UTAR	0.0222	2.84%	1	—

Portfolio Return: -0.82%

Table IX: Portfolio Composition (2025-04-03)

Ticker	Prediction Score	Weight	Cluster	Target Return
TRNFP	0.352582	20.81%	-1	15.78%
GMKN	0.215786	12.74%	-1	32.50%
TASBP	0.153386	9.05%	4	17.52%
NNSB	0.142174	8.39%	4	7.44%
YRSB	0.121512	7.17%	-1	2.66%
NNSBP	0.119371	7.05%	4	—
MISBP	0.111666	6.59%	0	—
RTSBP	0.100129	5.91%	0	—
VRSBP	0.095520	5.64%	0	—
SAGOP	0.054186	3.20%	5	—
TORS	0.046576	2.75%	5	—
VJGZP	0.034484	2.04%	1	—
KRSBP	0.030691	1.81%	3	—
VJGZ	0.027537	1.63%	1	—
KRSB	0.025005	1.48%	3	—
BISVP	0.025223	1.49%	2	—
JNOSP	0.020002	1.18%	2	—
EELT	0.018289	1.08%	2	—

Portfolio Return: 15.59%

Table X: Portfolio Composition (2025-05-05)

Ticker	Prediction Score	Weight	Cluster	Target Return
TRNFP	0.3325	22.06%	-1	14.79%
TASBP	0.2427	16.10%	0	30.54%
GMKN	0.2037	13.52%	-1	26.66%
NNSBP	0.1587	10.53%	0	23.09%
NNSB	0.1375	9.13%	0	28.11%
STSB	0.1333	8.85%	-1	—
MISBP	0.1045	6.93%	1	—
VRSBP	0.0996	6.61%	1	—
RTSBP	0.0947	6.28%	1	—

Portfolio Return: 25.16%

Table XI: Portfolio Composition (2025-06-06)

Ticker	**Prediction Score**	**Weight**	**Cluster**	**Target Return**
TRNFP	0.3660	23.81%	-1	6.12%
GMKN	0.2569	16.71%	-1	20.03%
TASBP	0.2011	13.08%	0	15.22%
NNSBP	0.1857	12.08%	0	4.96%
TNSE	0.1283	8.35%	0	3.23%
DZRD	0.1134	7.38%	-1	—
VRSBP	0.1025	6.67%	1	—
MISBP	0.0973	6.33%	1	—
RTSBP	0.0861	5.60%	1	—

Portfolio Return: 10.01%